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[22CSE140]

SECOND SEMESTER B.Tech EXAMINATION, JUNE - 2023

PYTHON PROGRAMMING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

PART - A

05 X 02=10

1.	**a.**	Compare python list and set.	L-1	CO1	**02**
	b.	Define python Dictionary.	L-1	CO2	**02**
	c.	Define constructor in python.	L-1	CO3	**02**
	d.	Define python package.	L-1	CO4	**02**
	e.	List the advantages of python Scipy Library	L-1	CO5	**02**
		PART - B **6 X 5=30**			
2.	**a.**	Write the syntax of python for loop statement and explain the working with an example.	L-2	CO1	**05**
		OR			
	b.	Write a python program to read a number from the keyboard and reverse it by using loop.	L-3	CO1	**05**
3.	**a.**	Explain python list and dictionary with suitable examples.	L-2	CO2	**05**
		OR			
	b.	Explain break and continue statements with suitable examples.	L-2	CO2	**05**
4.	**a.**	Explain constructor with suitable examples.	L-2	CO3	**05**
		OR			
	b.	Write a python program to display, append and delete elements from a list using class and methods.	L-3	CO3	**05**
5.	**a.**	Explain method overriding with suitable example.	L-2	CO4	**05**
		OR			
	b.	Write a python class Bank_Account with the attributes like Name, Balance and account number. Use the methods like deposit, withdraw and check_balance.	L-3	CO4	**05**
6.	**a.**	Compare NumPy arrays with Pythons List.	L-3	CO5	**05**

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		OR			
	b.	Write a python program for interchanging the axis of a NumPy array.	L-3	CO5	**05**
7.	**a.**	Write a python class for separating positive and negative numbers in a list. Use separate methods for positive and negative numbers.	L-3	CO5	**05**
		OR			
	b.	Write a python program to read N number of students details (ID, Name, Mobile and E-mail) as key:value pair and store the data in a CSV file. Read the input data from keyboard.	L-3	CO5	**05**
		PART - C **6 X 10=60**			
8.	**a.**	Explain any five string methods with suitable examples.	L-21	CO1	**10**
		OR			
	b.	Explain the conditional statements with syntax and proper examples.	L-21	CO1	**10**
9.	**a.**	Write a python class to collect 5 subject mark of N students. Make a three category of students with above 80, between 60 and 80 and below 60. Use separate methods for each operation. Use separate data structure for each.	L-3	CO2	**10**
		OR			
	b.	Explain recursion with suitable example.	L-3	CO2	**10**
10.	**a.**	Explain exception handling in python with suitable example.	L-2	CO3	**10**
		OR			
	b.	Add two complex number c= a+ bi and z= x+yi using magic methods in python. The code should read real and imaginary values from a user. Write a code using class , objects, and methods to solve the problem statement.	L-3	CO3	**10**
11.	**a.**	Explain inheritance with suitable example.	L-2	CO4	**10**
		OR			
	b.	Write a python package for performing a simple calculator of two numbers. The package having four methods like add(), sub(), mul() and div(). The input data is given from outside and the method should return the result.	L-3	CO4	**10**

12.	**a.**	Explain the vectorization in NumPy and demonstrate with python code.	L-3	CO5	**10**
		OR			
	b.	Explain about SciPy Cluster.	L-2	CO5	**10**
13.	**a.**	Write python class for online reservation system. The class should have the methods like reserve(), cancel(), waiting list(), display chart(). If the reservation chart is full, the data should be stored in waiting list and if any cancelation happens, it should immediately reflect in reservation chart and waiting list chart. Use appropriate data structure for storing the passenger information.	L-3	CO6	**10**
		OR			
	b.	Write a python class for analyzing the performance of 20 cricket players. Collect the data like "Player Name", "Match played", "Runs Scored", and "Wickets taken". Write class and methods for choosing 11 best players among this 20. Use separate methods for reading players data, analyzing players data and preparing the final 11 players. The input data should read from keyboard. Use Appropriate data structure for storing player's data.	L-3	CO6	**10**